ATMOSPHERIC SCIENCE

# Attributing 2019–2024 methane growth using TROPOMI satellite observations

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**Atmospheric methane concentrations increased at a rate of 0.7% year$^{-1}$ over 2019–2024, but the causes are unclear. We conducted an analytical inversion of bias-corrected TROPOspheric Monitoring Instrument (TROPOMI) satellite observations to quantify 2019–2024 annual mean methane emissions and hemispheric OH concentrations. We find that the methane rise from 2019 to 2024 reflects an approach to steady state between 2019 sources and sinks (59% of the rise), augmented by increasing emissions (25%) and decreasing OH concentrations (16%). Global emissions increased from 571 teragrams (Tg) per year in 2019 to 601 Tg per year in 2021 and back to 575 Tg per year in 2024. 2019–2024 emission decreases from oil/gas and rice were offset by increases from livestock and waste. East Africa and South America were most responsible for the 2019–2024 increase in emissions. The weaker methane rise over 2022–2024 was driven mostly by increasing OH concentrations rather than decreasing emissions.**



## INTRODUCTION

Methane ($CH_4$) is a highly potent greenhouse gas with ~80 times the global warming potential of carbon dioxide over a 20-year time span (*1*). The Global Methane Pledge signed by over 150 countries (*2*) commits to a collective reduction of anthropogenic methane emissions by 30% by 2030 relative to 2020 levels. The atmospheric methane concentration increased over 2007–2019 following a period of stabilization (*3*), then surged in 2020 and 2021 (*4*), and has sustained a high growth rate through 2024. The drivers for these trends are uncertain and could reflect changes in sources and/or sinks superimposed on an approach to steady state. The dominant natural source is wetlands, whereas anthropogenic sources include livestock, oil and gas facilities, landfills, coal mining, rice cultivation, and wastewater treatment. Ninety percent of the atmospheric methane sink is from oxidation by the hydroxyl radical (OH) in the troposphere, with additional minor sinks from oxidation by tropospheric chlorine radicals, oxidation in the stratosphere, and uptake by soils (*5*).

Atmospheric observations of methane and its isotopes have been used extensively to attribute methane trends, often reaching different conclusions. The post-2007 growth of global methane has been attributed to increased biogenic emissions from wetlands and livestock (*6*–*10*), oil and gas production (*11*, *12*), or a decline in OH (*13*, *14*). The record acceleration in 2020 has been attributed to increased tropical methane emissions (*15*, *16*), boreal wetland emissions (*17*), or COVID-19 lockdown-related decreases in OH (*18*, *19*). Microbial emissions from the wet tropics have been identified as major drivers of the sustained surge in 2021–2022 (*4*, *20*, *21*).

Satellite observations of dry-air column mixing ratios ($X_{CH4}$) in the shortwave infrared (SWIR) offer global continuous coverage to interpret trends in methane through inverse analyses. High-quality observations from the Greenhouse gases Observing SATellite (GOSAT) (2009–present) have been used in a number of studies (*21*–*25*) but are relatively sparse, with pixels separated by about 250 km. The more recent record from the TROPOspheric Monitoring Instrument (TROPOMI) (2018–present) provides daily global coverage at a nadir pixel resolution of 5.5 km by 7 km (7 km by 7 km before August 2019) (*26*) and has been used extensively in regional inversions (*27*–*32*) and in a 2018–2019 global inversion (*33*), but concerns over artifacts and regional biases (*34*) have limited application to global trends. The blended TROPOMI+GOSAT product of Balasus *et al.* (*35*) uses colocated GOSAT retrievals with machine learning to mitigate TROPOMI artifacts, thus combining TROPOMI data density and GOSAT data quality (*35*).

Here, we apply the blended TROPOMI+GOSAT satellite observations in a global inverse analysis of methane trends at 2° by 2.5° resolution from 2019 to 2024 to better understand the drivers behind the 2019–2024 methane trends, including trends in emissions from different sectors as well as trends in OH concentrations. We specifically examine whether progress has been made toward meeting the goals of the Global Methane Pledge.

## RESULTS

We conduct a global Bayesian analytical inversion of the blended TROPOMI+GOSAT observations (*35*), hereafter referred to as the TROPOMI observations, to quantify a state vector of annual surface fluxes at a 2° by 2.5° resolution and annual hemispheric OH concentrations for 2019–2024. The GEOS-Chem chemical transport model (*24*) is used to relate emissions to the TROPOMI observations. Starting from the best prior bottom-up estimates of this state vector in 2019, and error statistics for these prior estimates and for observations, the analytical inversion returns optimized (posterior) estimates of the state vector along with explicit error characterization, including a 27-member ensemble of solutions generated by varying inversion parameters from the base inversion. Our base inversion uses our best estimates of the inversion parameters, and the 27-member ensemble

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defines uncertainty ranges. Posterior estimates for each year from the base inversion are used as prior estimates for the next year following a Kalman filter approach, so that no bottom-up trend information is used in the inversion. The error characterization demonstrates the capability to separate emission and OH contributions to the methane trends through their different effects on the observed methane distribution (*24*, *36*, *37*). Further details are in the Materials and Methods section.

### Global drivers of the 2019–2024 methane trend

Figure 1 shows the global trend of atmospheric methane concentrations over the 2019–2024 period. The mean $X_{CH4}$ measured by TROPOMI increased at a mean rate of 12 parts per billion (ppb) year$^{-1}$ (0.7% year$^{-1}$) from 2019 to 2024, consistent with the trend reported by National Oceanic and Atmospheric Administration (NOAA) Marine Boundary Layer (MBL) surface monitoring sites in that same period. The GEOS-Chem simulation using our best posterior estimates of emissions and OH concentrations tracks the observed trends, and this is solely driven by the adjustments to emissions and OH concentrations with the Kalman filter. The posterior emissions and OH concentrations improve the GEOS-Chem model fit to the TROPOMI observations in each year relative to the prior estimates (see Materials and Methods; fig. S1). The bottom and top dashed lines in Fig. 1 show what the relaxation to steady state would have been if methane sources and sinks (mainly emissions and tropospheric OH) remained constant at posterior 2019 and posterior 2021 values, respectively, over the 2019–2024 period. The actual growth has been 57% higher than the 2019 relaxation, mainly driven by the acceleration over the 2020–2021 period. The global growth rate is now undershooting the 2021-condition steady state, implying either a decrease in emissions or increase in OH.

Figure 2A shows our best posterior estimates of 2019–2024 trends in global emissions and in the methane lifetime against oxidation by tropospheric OH, with uncertainties defined by the range of the 27-member ensemble. Analysis of the reduced averaging kernel matrix (see Materials and Methods) and the spread from the inversion ensemble demonstrate that the inversion can successfully separate the contributions of emissions and OH to the methane trend each year, as also shown in a previous work using GOSAT (*36*) and in an observing system simulation experiment for TROPOMI (*37*). This is because emissions and OH concentrations impose different patterns on the $X_{CH4}$ fields as expressed through the Jacobian matrix for the inversion.

We see from Fig. 2A that changing emissions and OH concentrations both contributed to the 2019–2024 methane trend. Global methane emissions increased from 571 Tg year$^{-1}$ in 2019 to a peak of 601 Tg year$^{-1}$ in 2021, before declining to 571 to 575 Tg year$^{-1}$ in 2022–2024 (Fig. 2A). Our best estimate for the methane lifetime against oxidation by tropospheric OH averages 11.1 years over 2019–2024, 6% longer than our prior estimate of 10.5 years (*38*) and within the estimate of 11.2 ± 1.3 years inferred by the methyl chloroform proxy (*39*). The interannual variability of the global OH concentration is 2%, consistent with the ±2% range from observations of methyl chloroform (*40*, *41*). The mean north-south interhemispheric OH ratio in the posterior estimate averages 1.01 for individual years (0.96 to 1.03 from the inversion ensemble), as compared to 1.07 in the prior estimate (*38*), and is more consistent with the observation-based estimate of 0.97 ± 0.12 (*42*). The uncertainty range from the inversion ensemble narrows as the inversion progresses because the spatial correction to the updated Kalman filter prior estimate decreases, reducing the sensitivity to the inversion parameters.

Tropospheric OH concentrations are sensitive to a range of chemical and meteorological variables in ways that are still poorly understood (*43*–*45*). Although some studies suggested that lower fossil fuel combustion emissions of nitrogen oxides ($NO_x$) from COVID-19 lockdowns in 2020 decreased tropospheric OH concentrations (*17*, *18*), our findings do not indicate a notable decline in OH from 2019 to 2020 (Fig. 2A). The OH concentration decreased by 2% from 2019 to 2022, and about a third of this can be attributed to the 2.3% increase in methane concentration over the period (Fig. 1), assuming a 0.31 feedback sensitivity (*46*). OH then increases from 2022 to 2024, indicating a dominant role of other factors.

The effects of changing emissions and OH on the 2019–2024 increase in methane concentrations are superimposed on a relaxation to steady state defined by 2019 conditions and taking place on the decadal scale of the methane lifetime. This superimposition can be expressed by a finite-difference mass balance equation for the annual growth rate $G_j = \Delta m / \Delta t$ (Tg year$^{-1}$) of methane atmospheric mass $m$ in year $j$ (*25*):

$$\begin{aligned} G_j &= E_j - k_j m_j - L_j \\ &= (E_0 + \Delta E_j) - (k_0 + \Delta k_j)(m_0 + \Delta m_j) - (L_0 + \Delta L_j) \\ &\approx (E_0 - k_0 m_0 - L_0 - k_0 \Delta m_j) + \Delta E_j - m_0 \Delta k_{OH,j} \end{aligned} \quad (1)$$

where $E$ denotes global emission, $k = k_{OH} + k_{other}$ is the global chemical loss rate constant that includes losses from tropospheric OH ($k_{OH}$) and other minor sinks including tropospheric chlorine radicals and the stratosphere ($k_{other}$), $L$ is the loss to soil, the subscript 0 refers to the initial 2019 conditions, and $\Delta$ refers to changes in year $j$ relative to 2019. The minor terms $m_0 \Delta k_{other,j}$, $\Delta k_j \Delta m_j$, and $\Delta L_j$ are neglected. The first term on the right-hand side of the final equation describes the relaxation to steady state from the initial 2019 conditions,

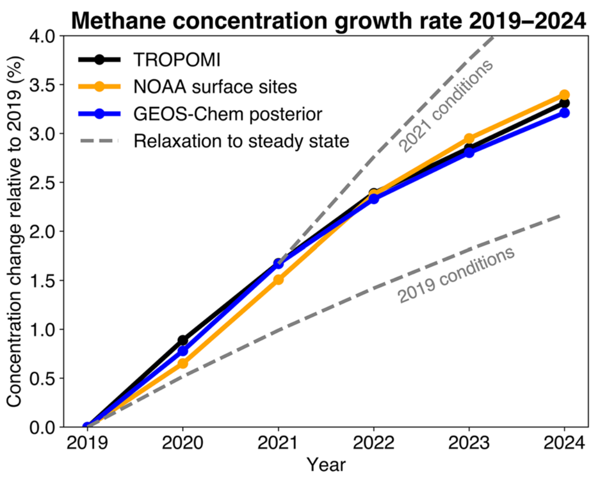


**Fig. 1. Trends in global annual mean methane concentrations from 2019 to 2024.** The NOAA trend is from globally averaged marine surface annual mean data (*87*). The TROPOMI trend is from blended TROPOMI+GOSAT observations (*35*). The GEOS-Chem posterior trend is from a simulation using our posterior estimates of methane emissions and OH concentrations for individual years, sampled at the locations of the TROPOMI observations and with the TROPOMI observation operator applied (*65*). The gray dashed lines show the evolution of atmospheric methane concentrations if sources and sinks had remained constant at either 2019 or 2021 values.

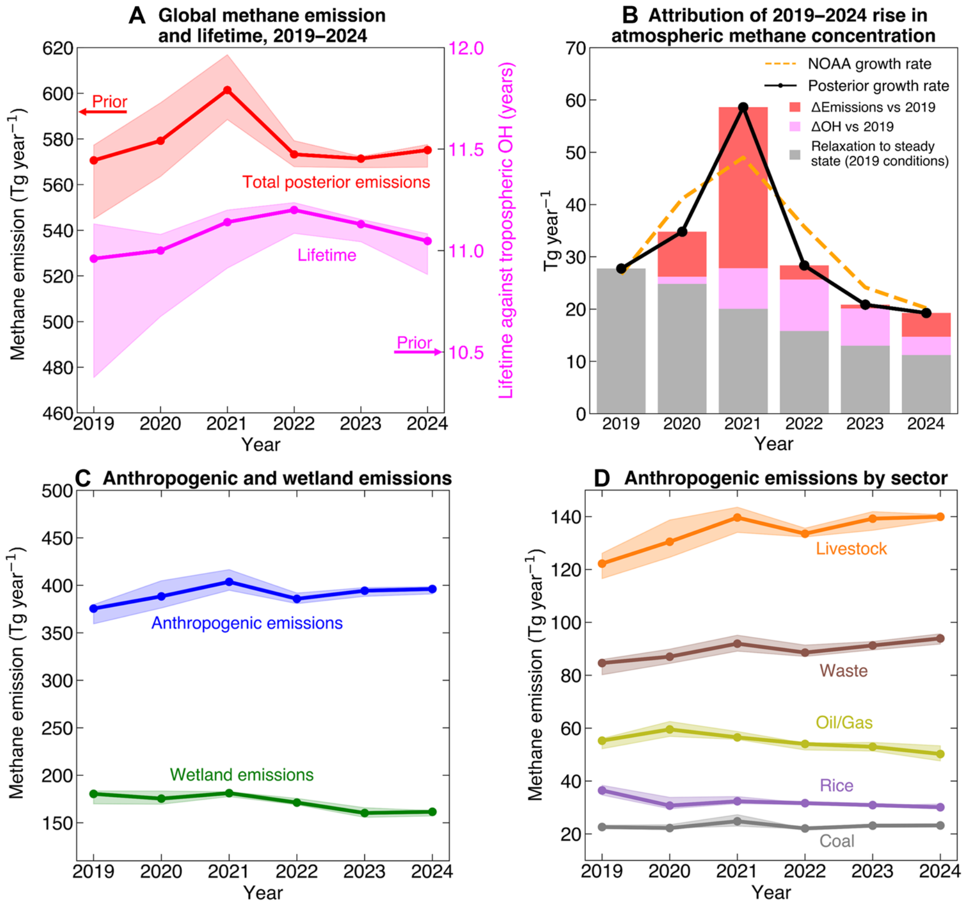


**Fig. 2. Contributions of global sources and sinks to the 2019–2024 methane trend as inferred from inversion of TROPOMI data.** (**A**) Posterior annual estimates of global methane emissions and methane lifetime against oxidation by tropospheric OH. Solid lines indicate results from our base inversion, and shaded areas represent the range from a 27-member inversion ensemble with varying inversion parameters (see Materials and Methods). (**B**) Attribution of the 2019–2024 year-to-year rise in atmospheric methane to relaxation to steady state from 2019 conditions ($E_0 - k_0 m_0 - L_0 - k_0 \Delta m_j$ term in Eq. 1), changes in methane emissions relative to 2019 ($\Delta E_j$), and changes in tropospheric OH concentrations relative to 2019 ($m_0 \Delta k_{OH,j}$). The orange dashed line shows the annual atmospheric methane growth rates as reported by NOAA (*87*) from the deseasonalized trends at marine surface sites and using a conversion factor of 2.77 Tg methane per ppb (*88*). (**C**) Posterior annual estimates of global anthropogenic emissions and wetland emissions, with ranges from the inversion ensemble. (**D**) Posterior annual estimates of global emissions from major anthropogenic sectors including livestock, waste (landfills and wastewater treatment), oil and gas, rice agriculture, and coal mining, with ranges from the inversion ensemble. Note the different *y*-axis scales between panels.

the second term describes the perturbation to emissions relative to the initial conditions, and the third term describes the perturbation to OH.

Figure 2B shows the contributions to the 2019–2024 growth of atmospheric methane from the different terms in Eq. 1 and for individual years. We find that relaxation from the 2019 conditions contributes 59% of the 2019–2024 trend; this means that if emissions and OH concentrations remained constant at 2019 values, the rise in atmospheric methane from 2019 to 2024 would have been 59% of its actual value. Rising emissions over 2019–2024 contributed 47 Tg or 25% of the rise (sum of $\Delta E_j$ for all years) and decreasing OH contributed 30 Tg or 16% of the rise (sum of $m_0 \Delta k_{OH,j}$ for all years).

The contributions to year-to-year trends in atmospheric methane from changes in emissions and OH can also be inferred from Fig. 2B. The 2020–2021 methane surge beyond the relaxation to steady state from 2019 conditions is due 81% to rising emissions and 19% to decreasing OH. The lower growth rate in 2022 is driven by decreasing emissions, with some offset from decreasing OH. The continued decrease in growth rate in 2023 and 2024 is mostly due to increasing OH.

By mapping the 2019–2024 trends on the 2° by 2.5° grid to the prior bottom-up distribution of 2019 emissions on that same grid, we find that anthropogenic sources drove the 2019–2021 emission increase, rising from 375 Tg year$^{-1}$ in 2019 to 404 Tg year$^{-1}$ in 2021, then decreasing in 2022, before increasing slightly again to 396 Tg year$^{-1}$ in 2024 (Fig. 2C). Wetland emissions were flat in 2019–2021 and then decreased from 181 Tg year$^{-1}$ in 2021 to 162 Tg year$^{-1}$ in 2024, mostly due to decreasing Amazon emissions, which offset an increase in Africa. The flat 2019–2021 global wetland emissions are consistent with a previous inversion of GOSAT satellite observations (*21*). The optimized soil sink averages 35 Tg year$^{-1}$ across all 6 years, showing minimal difference from the prior estimate and no trend.

Figure 2D further breaks down the global anthropogenic emission trend into the contributions from individual sectors. Between 2019 and 2024, emissions from the oil/gas sector decreased (−5 Tg year$^{-1}$, −9%) and so did rice emissions (−6 Tg year$^{-1}$, −17%), but this was offset by increases in emissions from livestock (18 Tg year$^{-1}$, 15%) and waste (9 Tg year$^{-1}$, 11%), whereas coal emissions stayed flat. Oil/gas emissions peaked at 60 Tg year$^{-1}$ in 2020 and have fallen by 16% since then. The 2021 peak in methane emissions is primarily driven by a 7% increase in emissions from livestock between 2020 and 2021, with additional contributions from waste (6% increase) and wetlands (3% increase). This aligns with atmospheric observations of the methane $^{13}$C:$^{12}$C ratio, which attribute the 2020–2022 methane surge to a biogenic source (*20*). However, bottom-up uncertainties in the spatial distributions of wetlands and livestock lead to some uncertainty in sectoral attribution of our inversion results (*47*).

## Regional drivers of the 2019–2024 methane trend

Figure 3A shows the global distribution of methane surface fluxes in our base posterior estimate for 2024. Global mean methane emissions in 2024 are 575 Tg year$^{-1}$ (568 to 578 Tg year$^{-1}$ range from the inversion ensemble). We find that wetlands are the largest total source of methane (28%), followed by livestock (25%), waste (16%), and oil and gas (9%). Averaging kernel sensitivities, which measure the ability of the observations to quantify individual elements of the state vector independently from the prior estimates, are near unity in regions with high emissions and dense TROPOMI observations (fig. S2). The degrees of freedom for signal (DOFS) on emissions, representing the number of independent pieces of information in the inversion and computed as the trace of the averaging kernel matrix (global sum of averaging kernel sensitivities), is 295 for 2024. It ranges from 256 to 426 for individual years, reflecting changes in satellite coverage. The satellite data provide strong constraints on emissions in major anthropogenic source regions such as India and East Asia. Averaging kernel sensitivities are low over the wet tropics including the Amazon and Southeast Asia, where observations are sparse due to cloud cover and/or dark surfaces (fig. S2). Averaging kernel sensitivities are also low in regions of low prior emissions, such as northern Australia and the Sahara.

The global distribution of 2019–2024 methane emission trends, fitted by ordinary linear regressions to the annual posterior emissions for individual 2° by 2.5° grid cells, is shown in Fig. 3B. Trends by sector are shown in fig. S3. East Africa is the fastest growing region (1.5 Tg year$^{-2}$; 1.4 to 1.7 Tg year$^{-2}$ for the 27-member inversion ensemble), followed by South America (1.3 Tg year$^{-2}$; −0.4 to 2.6 Tg year$^{-2}$) and Europe (1.1 Tg year$^{-2}$; 0.9 to 1.5 Tg year$^{-2}$). Emissions across Africa increased from 79 Tg year$^{-1}$ in 2019 to 86 Tg year$^{-1}$ in 2024 (0.9 Tg year$^{-2}$; 0.8 to 1.3 Tg year$^{-2}$) and account for 28% of the 2019–2021 emissions surge. This is consistent with Qu *et al.* (*21*), who found from GOSAT data that Africa drove 30% of the 2020–2022 surge. Pendergrass *et al.* (*47*) also attribute the 2020 surge to emissions from sub-Saharan Africa using TROPOMI data. The increase in East Africa is mostly due to wetland and livestock emissions, and there is uncertainty in separating the two as previously noted. Emissions increased by 0.2 Tg year$^{-2}$ (0.2 to 0.3 Tg year$^{-2}$) in the Sudd wetlands of South Sudan and their surrounding catchment areas (including Lake Victoria) over 2019–2024, with a



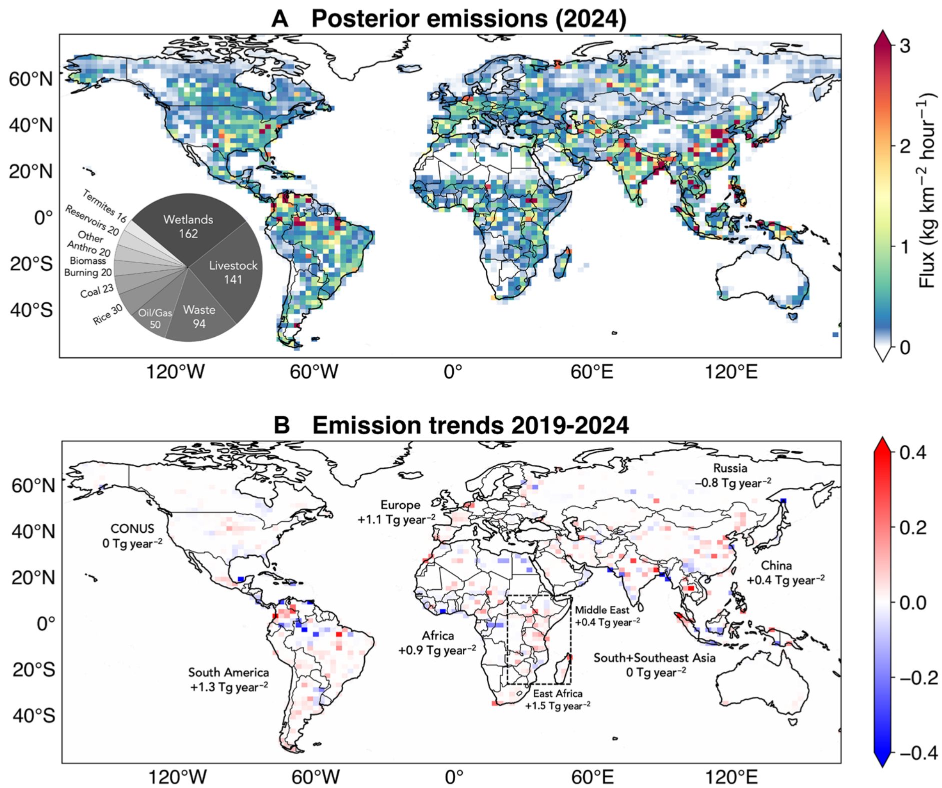


**Fig. 3. Regional distribution of methane emissions and 2019–2024 trends from inversion of TROPOMI data.** (**A**) 2024 mean posterior methane emissions from the base inversion. Pie chart shows the sectoral attribution of global emissions (Tg year$^{-1}$) in 2024. Ocean fluxes are assumed to be zero. (**B**) 2019–2024 methane emission trends fitted by ordinary linear regressions to the annual posterior emissions in each 2° by 2.5° grid cell. Only significant trends ($P < 0.05$) are shown. Total growth rates for individual geographical regions are shown in the inset.

peak of 5 Tg year$^{-1}$ in 2022. This is consistent with positive emission trends over the region detected in earlier years from GOSAT inversions (*24*, *48*, *49*), likely due to inflow from the White Nile and Sobat rivers leading to increased inundation extent and water height since 2017 (*48*, *50*, *51*).

On the basis of the sectoral distribution of our prior inventory, livestock emissions in East Africa increased 0.6 Tg year$^{-2}$ (0.5 to 0.8 Tg year$^{-2}$) over 2019–2024. Zhang *et al.* (*25*) also found that livestock emissions increased rapidly in East Africa and contributed to the overall rise in emissions from 2010 to 2018. Although livestock population trends reported by the Food and Agriculture Organization (FAO) suggest a smaller increase in sub-Saharan Africa (*47*), FAO statistics are known to be uncertain given challenges in census collection in Africa (*52*). Improved sectoral attribution in our spatially resolved inversion results is contingent on better bottom-up information on the spatial distribution of emissions. The LPJ-MERRA2 wetlands inventory used here as prior estimate relies on a precipitation-based hydrological model to simulate inundation extent and this may underrepresent inundation from river flooding, particularly in East Africa (*51*). East Africa has high and increasing levels of inundation observed by the Gravity Recovery and Climate Experiment Follow-On (GRACE-FO) twin satellites, but this is not reflected in LPJ-MERRA2 or other wetlands inventories (*47*). Inundation as measured by GRACE-FO correlates with methane emission seasonality and increases in sub-Saharan Africa (*47*). However, GRACE-FO measures water storage anomalies rather than absolute storage, so relative trends in inundation cannot be directly computed to reconcile with emission trends. It is thus unclear whether inundation as observed by GRACE-FO is sufficient to explain the rapid increases in emissions from East Africa. Correlated $NH_3$ trends from the Infrared Atmospheric Sounding Interferometer (IASI) satellite instrument have been used to attribute methane emissions to the livestock sector in tropical regions (*53*), but interpretation is complicated by nonlivestock sources of $NH_3$, uptake of $NH_3$ by aerosols, and vertical dependence of the $NH_3$ retrieval sensitivity. The IASI observations show decreasing ammonia concentrations in East Africa over 2019–2024 (fig. S4), which does not support an increasing livestock source (*54*).

We find that South America is also a major contributor to the 2019–2024 methane rise, with total emissions increasing from 100 Tg year$^{-1}$ in 2019 to 105 Tg year$^{-1}$ in 2024 (1.3 Tg year$^{-2}$; −0.4 to 2.6 Tg year$^{-2}$). The trend is driven by an increase of 0.8 Tg year$^{-2}$ (0.6 to 0.8 Tg year$^{-2}$) in fuel (oil/gas, mostly in Venezuela and Colombia) and waste emissions. Wetland and livestock emissions did not change significantly. At the country level, emissions increased in Colombia (0.2 Tg year$^{-2}$; 0.2 to 0.4 Tg year$^{-2}$) and decreased in Brazil (−0.3 Tg year$^{-2}$; −0.8 to 0.1 Tg year$^{-2}$). Rising anthropogenic emissions in Colombia were previously found in a global inversion of GOSAT and surface observations over 2009–2020 (*55*). Emissions decreased in the Amazon basin (−2.7 Tg year$^{-2}$; −2.1 to −3.5 Tg year$^{-2}$), reflecting record drought conditions in the region in 2023 (*56*). There are weak positive error correlations ($|r| < 0.25$) between South America and Oceania (Eq. 6 and fig. S5) for reasons that are unclear but might reflect some commonality in response to errors in OH.

We find that emissions in Europe increased from 19 Tg year$^{-1}$ in 2019 to 26 Tg year$^{-1}$ in 2024, due to livestock and waste, but the spatial overlap of sources on the 2° by 2.5° grid of our inversion limits the capability for sectoral attribution. Emissions in China increased by 0.4 Tg year$^{-2}$ (0.1 to 0.6 Tg year$^{-2}$) over 2019–2024, with rising emissions in northern China offsetting decreases in southern China as coal mining activities shifted northward (*57*). This is consistent with the 2010–2022 trend reported by Qu *et al.* (*21*) from GOSAT data. There are weak negative error correlations ($|r| < 0.25$) between adjacent regions such as China and Japan/Korea (Eq. 6 and fig. S5), pointing to emissions that cannot be fully differentiated. We do not find significant 2019–2024 emission trends in the contiguous US or Canada. In South and Southeast Asia, strong error correlations between livestock and waste and overlapping anthropogenic and wetland sources again limit our ability to separate sectoral emissions (figs. S5 and S6).

Boreal wetland emissions decreased over 2019–2024 by 0.2 Tg year$^{-2}$ (−0.1 to −0.2 Tg year$^{-2}$) in the Hudson Bay Lowlands and 0.3 Tg year$^{-2}$ (−0.3 to −0.5 Tg year$^{-2}$) in the West Siberian Lowlands. Boreal ecosystems in Russia and Canada have experienced persistent drought stress as evidenced by declining water table depth (*58*) and declining water thickness observed by GRACE-FO (*47*). Increased evapotranspiration leading to drier wetland soils may contribute to the downward trend in emissions (*59*). Previous studies have found decreasing wetland emissions in boreal North America over 2000–2019 (*60*, *61*).

## DISCUSSION

We find from analytical inversion of TROPOMI satellite observations that the 2019–2024 methane rise reflects contributions from a decadal approach to steady state driven by 2019 conditions (59%), increasing emissions (25%), and decreasing OH sink (16%). The peak methane growth rate in 2021 reflected a surge in emissions, which increased from 571 Tg year$^{-1}$ in 2019 to 601 Tg year$^{-1}$ in 2021 but have decreased back to 571 to 575 Tg year$^{-1}$ since then. OH concentrations decreased by 2% from 2019 to 2022 and have since partly recovered. The growth rate decreased over 2022–2024, but we find that this is largely driven by increasing OH. East Africa has the highest emissions growth rate from 2019 to 2024, followed by South America and Europe. Trends in the US and Canada were flat. Oil/gas emissions decreased by 9% during 2019–2024, with notable declines in the Middle East and North Africa (fig. S3), indicating initial progress toward the goal of the Global Methane Pledge (*62*). Rice emissions also fell by 17%, but these decreases were more than offset by increases in emissions from livestock (15%) and waste (11%). Strong action to decrease emissions from livestock and waste will be essential to fulfill the goal of the Pledge.

Given the extent of spatial overlap in anthropogenic and wetland source regions of Africa and South America (fig. S6), as well as the limited observational coverage due to cloudiness, our ability to separate these tropical sectors remains limited. Increasing the resolution of the inversion could reduce uncertainty in sectoral attribution but only with correspondingly dense observations, which TROPOMI does not achieve in the tropics. Increasing pixel resolution for better cloud clearing, as with the proposed Carbon-I satellite instrument (*63*), would greatly increase the information content through higher observational density in wetland regions of the tropics and northern high latitudes (fig. S7). This needs to be combined with improved bottom-up estimates of the spatial distribution of livestock and wetland emissions for attribution of inversion results.

## MATERIALS AND METHODS

We use the Integrated Methane Inversion (IMI) version 2.0 open-access computing tool to perform annual inversions for 2019–2024 (*64*). The IMI was originally designed for regional inversions (*65*), and this work represents the global capability implemented in version 2.0. The inversion is applied to the 2019–2024 TROPOMI versions 02.04–02.08 operational retrievals of $X_{CH4}$ (*66*), corrected for artifacts using collocated GOSAT retrievals and machine learning (*35*). We only use retrievals with a quality assurance value equal to 1, exclude glint data over water because of persistent artifacts (*35*), and exclude permanently snow- or ice-covered scenes based on TROPOMI surface classification. TROPOMI data are missing for 26 July to 23 August in 2022, 10 to 30 August in 2023 (but retrievals begin to fail on 26 July), and 29 May to 3 June, 11 to 18 July, 26 to 31 July, 26 to 29 September, and 3 to 7 November in 2024 due to missing Visible Infrared Imaging Radiometer Suite (VIIRS) cloud data. Some adjacent days to these periods have anomalous values that we also filter out. The inversion ingests a total of 610 million successful TROPOMI retrievals for the 2019–2024 6-year period.

### Prior estimates and Kalman filter

The inversion optimizes an annual mean global state vector consisting of 4111 2° by 2.5° surface flux elements (land-containing grid cells) plus hemispheric tropospheric OH concentrations (two elements), for a total of 4113 state vector elements each year. Initial prior estimates for 2019 are from gridded bottom-up inventories including EDGARv7 as global anthropogenic default (*67*), superseded with the monthly Global Rice Paddy Inventory (GRPI) for rice emissions (*68*), the Global Fuel Exploitation Inventory (GFEI) version 2.0 for fuel exploitation (*69*), the EPA Greenhouse Gases Inventory (GHGI) version 2.0 Express Extension for the contiguous US (*70*), and regional inventories for Mexico (*71*) and Canada (*72*). Hydroelectric reservoir methane emissions are added with the ResME inventory (*73*). Seasonal scaling factors are applied locally to manure emissions (*70*). For wetland emissions, we use monthly estimates from the process-based LPJ-EOSIM model driven by MERRA-2 meteorology (hereafter LPJ-MERRA2) (*74*, *75*), which better captures observed methane seasonality in the Northern Hemisphere compared to the WetCHARTs inventory ensemble previously used in global inversion studies (*76*). Biomass burning emissions are at daily resolution from GFED4 (*77*), termite emissions are from Fung *et al.* (*78*), and geological seeps are from Etiope *et al.* (*79*) scaled to the global annual value provided by Hmiel *et al.* (*80*). Global soil uptake is from a 1990–2009 climatology of the Soil Methanotrophy Model (MeMo v1.0) (*81*).

The initial prior estimate for methane loss from oxidation by tropospheric OH is computed from archived three-dimensional (3D) monthly OH fields generated by a GEOS-Chem full-chemistry simulation (*38*). This yields a methane lifetime of 10.5 years against oxidation by tropospheric OH. Other minor chemical sinks are not optimized in the inversion.

The 6-year 2019–2024 inversion is conducted as a suboptimal Kalman filter, where the posterior estimate of the state vector for a given year is used as prior estimate for the next year. The same prior error estimates are used for all 6 years of the inversion, so that error shrinkage in the posterior estimate is not applied to the prior estimate for the next year. Prior error SDs for the surface fluxes on the 2° by 2.5° grid are taken to be 50%, and prior error SDs for the hemispheric annual mean OH concentrations are taken to be 10% (*36*). We conduct a 1-year burn-in for 2019 starting from the initial prior estimates and then use those results as the prior estimate for a second 2019 inversion with the same observations. This is to ensure that the 2019–2024 global trend is not biased by influence from the initial prior estimates. We verified that the posterior estimates for 2019 were not substantially different between the burn-in year and the second 2019 inversion.

### Analytical inversion

We use the GEOS-Chem (v14.4.1) 3D chemical transport model as the forward model to relate surface emissions to column methane concentrations. GEOS-Chem simulates methane concentration fields globally at 2° by 2.5° horizontal resolution with 47 vertical layers, including all sources and sinks (*24*), and is driven by NASA GEOS-FP meteorological fields. The initial conditions for the inversion on 1 January 2019 are taken from an archive of smoothed TROPOMI fields, combined with a long-term GEOS-Chem simulation from a properly initialized stratosphere (*82*). This achieves an unbiased representation of $X_{CH4}$ relative to TROPOMI including the partitioning of the column between the troposphere and stratosphere (*64*).

Bayesian optimization of the state vector $\mathbf{x}$, assuming normal error probability density functions for the prior estimate $\mathbf{x_A}$ (error covariance matrix $\mathbf{S_A}$) and for the observation vector $\mathbf{y}$ (error covariance matrix $\mathbf{S_o}$), minimizes the cost function $J(\mathbf{x})$(*83*)

$$J(\mathbf{x}) = \left(\mathbf{x}-\mathbf{x_A}\right)^{T}\mathbf{S_A}^{-1}\left(\mathbf{x}-\mathbf{x_A}\right) + \gamma(\mathbf{y}-\mathrm{Kx})^{T}\mathbf{S_O}^{-1}(\mathbf{y}-\mathrm{Kx}) \quad (2)$$

where $\mathbf{K} = \partial\mathbf{y}/\partial\mathbf{x}$ is the Jacobian matrix representing the forward model sensitivity of the observations to a perturbation in the state vector, and $\gamma$ is a regularization factor to prevent overfitting. The observation vector averages TROPOMI retrievals over individual 2° by 2.5° grid cells and orbits to yield super-observations accounting for error correlations (*28*).

The prior error covariance matrix $\mathbf{S_A}$ and the observational error covariance matrix $\mathbf{S_O}$ are both taken as diagonal. The observational error SD $\sigma_o$ includes contributions from retrieval and model transport errors. We estimate it for individual super-observations as a function of the number $P$ of individual retrievals averaged into the super-observation, following the method of Chen *et al.* (*28*). This builds on the observational error SD for individual retrievals using the residual error method (*84*) but accounts for the perfect model transport error correlation as well as the partial retrieval error correlation for retrievals averaged into super-observations. Chen *et al.* (*28*) found that, for 25 km by 25 km grid cells, $\sigma_o$ approached a transport model error asymptote at large values of $P$, but here we do not find such an asymptote, possibly because model transport errors are smaller on the 2° by 2.5° grid. Instead, we fit a spline function to $\sigma_o$ versus $P$ (fig. S8), which yields a mean $\sigma_o$ of 5.8 ppb for the super-observations. Error correlation in the observations likely extends to adjacent grid cells and successive orbits, but this is not resolved here; the inversion therefore requires a regularization factor to avoid overfit. We select $\gamma = 0.02$ following (*85*) to match the expected chi-square value for the sum of prior terms in the cost function. Using $\gamma = 1$ would result in excessive corrections to the prior estimates relative to the prior error variance, manifested as dipoles in the inversion solution (i.e., checkerboarding) that are symptomatic of overfit (fig. S9).

The Jacobian matrix $\mathbf{K}$ is explicitly constructed through GEOS-Chem simulations perturbing individual state vector elements (*64*). The forward model relationship between methane emissions and $X_{CH4}$ is strictly linear, and the relationship between OH concentrations and $X_{CH4}$ is only weakly nonlinear on the 1-year timescale of the inversion, so that $\mathbf{K}$ fully describes the forward model. We tested the effect of OH nonlinearity by computing $\mathbf{K}$ iteratively and found it to be negligible. To balance the influence of emissions and OH in the cost function, we increase the weight of the two hemispheric OH terms following Maasakkers *et al.* (*24*).

Minimizing the cost function in Eq. 2 yields an analytical solution for the posterior solution $\hat{\mathbf{x}}$

$$\hat{\mathbf{x}} = \mathbf{x_A} + \left(\gamma \mathbf{K}^T \mathbf{S_O}^{-1} \mathbf{K} + \mathbf{S_A}^{-1}\right)^{-1} \gamma \mathbf{K}^T \mathbf{S_O}^{-1} \left(\mathbf{y} - \mathbf{K} \mathbf{x_A}\right) \tag{3}$$

The posterior error covariance matrix $\hat{\mathbf{S}}$ characterizes the error on $\hat{\mathbf{x}}$

$$\hat{\mathbf{S}} = \left(\gamma \mathbf{K}^T \mathbf{S_O}^{-1} \mathbf{K} + \mathbf{S_A}^{-1}\right)^{-1} \tag{4}$$

Errors in inversion parameters are not included in $\hat{\mathbf{S}}$, so we conduct an inversion ensemble for individual years to better characterize uncertainty in the posterior estimates (*27*). The ensemble varies the prior emission error SD (20, 50, and 70%), the prior hemispheric OH error SD (5, 10, and 15%), and the regularization parameter $\gamma$ (0.005, 0.02, and 0.05), with the midpoints representing the conditions of the base inversion. These sensitivity inversions test different parameters used in previous studies (*21*, *24*, *25*). The resulting spread with 27 ensemble members is larger than represented by $\hat{\mathbf{S}}$ (by a factor of 5 on average) and is used here to more conservatively characterize errors in our results.

Comparing $\hat{\mathbf{S}}$ to $\mathbf{S_A}$ quantifies the information content of the inversion. The averaging kernel matrix $\mathbf{A} = \partial \hat{\mathbf{x}} / \partial \mathbf{x} = \mathbf{I} - \hat{\mathbf{S}} \mathbf{S_A}^{-1}$ describes the sensitivity of the solution to the true state. The diagonal elements of $\mathbf{A}$ measure the ability of the observations to constrain individual state vector elements independently from the prior estimate. The trace of $\mathbf{A}$ gives the DOFS, or the number of independent pieces of information on the state vector gained from the observations (*83*). We evaluated the base inversion results for individual 2019–2024 years by conducting posterior GEOS-Chem simulations driven by posterior estimates of methane emissions and hemispheric OH fields to compare to prior GEOS-Chem simulations driven by the prior estimates. We find that the prior GEOS-Chem simulation for 2019 (starting from the initial prior estimate before the burn-in) has a mean bias of only 0.43 ppb relative to the mean TROPOMI observations for that year, with a root mean square error (RMSE) relative to annual mean TROPOMI concentrations on the 2° by 2.5° grid of 9.36 ppb. The posterior simulation for 2019 improves the RMSE to 7.98 ppb (fig. S1). Posterior simulations for succeeding years maintain a low bias compared to TROPOMI observations for those years (−0.25 to 1.06 ppb) and a low RMSE (7.67 to 8.56 ppb).

### Separating contributions of sectoral emissions and OH concentrations to methane trends

The posterior solution $\hat{\mathbf{x}}_{4113\times1}$ on the 2° by 2.5° grid can be aggregated spatially and/or by sector to a smaller dimension $n$ ($\hat{\mathbf{x}}_{\mathrm{red}\ n\times1}$) via a summation matrix $\mathbf{W}$

$$\hat{\mathbf{x}}_{\mathrm{red}} = \mathbf{W} \hat{\mathbf{x}} \tag{5}$$

The reduced posterior error covariance and averaging kernel matrix are

$$\hat{\mathbf{S}}_{\mathrm{red}} = \mathbf{W} \hat{\mathbf{S}} \mathbf{W}^T \tag{6}$$

$$\mathbf{A}_{\mathrm{red}} = \mathbf{W} \mathbf{A} \mathbf{W}^* \tag{7}$$

where $\mathbf{W}^* = \mathbf{W}^T \left(\mathbf{W} \mathbf{W}^T\right)^{-1}$ is the Moore-Penrose pseudoinverse (*86*).

We aggregate posterior emissions by sector based on the relative contributions of that sector to total prior emissions in individual 2° by 2.5° grid cells and by region based on geographical boundaries (fig. S5) (*25*). The inversion optimizes surface fluxes, which may, in principle, be negative because of the soil sink. As part of our sectoral attribution of surface fluxes, we disaggregate the optimized flux from the inversion into its emission and sink components. We use the quadratic equation (Eq. 8) to convert the posterior/prior flux ratio returned by the inversion to a separate correction factor for emissions ($C$) and the soil sink ($1/C$), assuming that the correction factor $C$ to the prior flux estimate reflects the same relative errors for the emissions and the soil sink. The prior surface flux estimate $\mathbf{x_A}$ includes a contribution $\alpha_{\mathbf{A}}$ for the soil sink. We solve for $C$ for each grid cell $i$ as the root of the quadratic equation

$$\hat{x}_i = \left(x_{A,i} - \alpha_{A,i}\right) * C - \frac{\alpha_{A,i}}{C} \tag{8}$$

The ability of the global inversion to separate the contributions of emissions and OH concentrations in the global methane budget has been demonstrated previously in an observation system simulation experiment for TROPOMI observations (*37*) and in actual inversions using the sparser GOSAT observations (*21*, *24*, *25*, *36*). Here, we evaluate the ability of our inversion to separate global emissions (E) and global mean OH ($\left[\overline{\mathrm{OH}}\right]$) by constructing the 2 by 2 matrix $\mathbf{A}_{\mathrm{red}}$, following Penn *et al.* (*36*). The diagonal elements $a_{\mathrm{red}\ 1,1} = \partial \hat{\mathrm{E}} / \partial \mathrm{E}$ and $a_{\mathrm{red}\ 2,2} = \partial \widehat{\left[\overline{\mathrm{OH}}\right]} / \partial \left[\overline{\mathrm{OH}}\right]$ quantify the sensitivity of posterior emissions to true emissions and the sensitivity of posterior OH to a change in true OH, respectively (Eq. 9). A sensitivity of 1 indicates perfect characterization of the reduced state vector element of interest. The off-diagonal elements quantify cross-sensitivities (i.e., aliasing) between emissions and OH. We find that the diagonal averaging elements for each year are near unity with negligible aliasing from emissions, demonstrating that OH can be optimized independently of emissions each year. The 2024 reduced averaging kernel matrix in Eq. 9 is a representative example

$$\mathbf{A}_{\mathrm{red}} = \begin{bmatrix} \partial \hat{\mathrm{E}} / \partial \mathrm{E} & \partial \hat{\mathrm{E}} / \partial \left[\overline{\mathrm{OH}}\right] \\ \partial \widehat{\left[\overline{\mathrm{OH}}\right]} / \partial \mathrm{E} & \partial \widehat{\left[\overline{\mathrm{OH}}\right]} / \partial \left[\overline{\mathrm{OH}}\right] \end{bmatrix} = \begin{bmatrix} 0.90 & -0.01 \\ -0.10 & 0.99 \end{bmatrix} \tag{9}$$

We also find in our inversion ensemble that separation between emissions and OH concentrations is consistent across the range of inversion parameters (Fig. 2A).

## Supplementary Materials

**This PDF file includes:**

Figs. S1 to S9

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# Attributing 2019–2024 methane growth using TROPOMI satellite observations

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